

SWE 4504

SSH PROTOCOL

Task 1

Make a server and a client. The server will give command to the client and the client will execute them and send the outputs of those commands to the server. These server and client will utilize SSH protocol to secure their communication (remote login and command-line execution). First, we need to follow some steps to build the necessary environment.

1. Make a virtual environment using : `python -m venv ssh`
2. Activate the ssh virtual environment using:
 - a. For linux : `source ssh/bin/activate`
 - b. For windows : `ssh\Scripts\activate`
3. Install paramiko using : `pip install paramiko`
4. Create a file called `test_rsa.key` in the current working directory and copy paste the content of this [file](#) to the created file `test_rsa.key`
5. Create `ssh_rcmd.py` and `ssh_server.py` utilizing the black hat python book or slide 6.
6. Do the above steps in both of your systems (Your host os Windows and guest os Kali)
7. Execute `ssh_server.py` (server) in kali first then run `ssh_rcmd.py` (client) in windows. Then, give some commands to client like (`whoami/ipconfig`) and show the outputs in server.
8. You may face some errors while using `getpass` module, you can discard that and use vanilla input method of python for user and password. Another error is session closed error. Always remember to close your session outside the loop of communication.

Task 2

You need to implement the bottom right scenario called SSH Tunneling. To implement this you need to follow the code provided in the slide for SSH Tunneling and some steps.

1. You need to be in the ssh virtual environment created in task 1.
2. Create `ssh_tunneling.py` file using the code provided in the slide 6.
3. Give your kali username and password in `ssh_user` and `ssh_password` variable inside `ssh_tunneling.py` file.
4. Initiate a web server at 8080 port using python executing the following command in a kali linux terminal:

```
python -m http.server 8080
```

5. Next execute `ssh_tunneling.py` in a second terminal using : `python ssh_tunneling.py`
6. Open a third terminal and execute the following command to access the web server:

```
curl 127.0.0.1:9090
```

7. You should see some sort of output in the first terminal.
8. If not there is some error you need to solve first.
9. And remember all the terminals should have ssh venv activated.

